

Miniaturized antenna for cognitive communication in the IoT applications.

Mohammed Zakarya BABA-AHMED
University of Tlemcen, Faculty of
technology, Laboratory of
Telecommunication (LTT)
University of Chlef, Faculty of
technology
Tlemcen, Algeria
zaki.babaahmed@gmail.com

Mohammed Amin RABAH
Algerian Space Agency_Satellite
Development Center
Laboratory of Telecommunication
(LTT)
Tlemcen, Algeria
arabah6@gmail.com

Rahma Djaouda TALEB
University of Tlemcen, Faculty of
technology
Laboratory of Telecommunication
(LTT)
Tlemcen, Algeria
rahmadjaouda.taleb@gmail.com

Soheyb Abderrahmane BELAHSEN
University of Tlemcen, Faculty of
technology
Laboratory of Telecommunication
(LTT)
Tlemcen, Algeria
sohaibbelahcen@gmail.com

Malika MESMOUDI
University of Tlemcen, Faculty of
technology
Laboratory of Telecommunication
(LTT)
Tlemcen, Algeria
malika.mesmoudi1@gmail.com

Abstract—In this paper, we present an in-depth study on the design and simulation of a miniaturized antenna specifically designed for cognitive communications in IoT applications. Utilizing CST software, we conducted simulations and optimizations on the antenna's performance, targeting a central frequency of 5.8 GHz—a standard in both wireless networks and IoT communications.

Our design methodology follows an iterative approach, wherein we fine-tune antenna parameters to attain optimal performance in gain, directivity, reflection coefficient, and efficiency. We additionally addressed miniaturization constraints by implementing geometric modifications in the ground plane and patch to facilitate seamless integration into IoT devices.

The simulation results unequivocally demonstrate the miniaturized antenna's outstanding performance, delivering both a commendable gain and adequate directivity, all the while maintaining a compact form factor with a 27% reduction in size. These attributes position it as an excellent choice for IoT devices where space and power consumption are critical considerations.

Keywords—antenna, cognitive radio, CST, Iot, miniaturization.

I. INTRODUCTION

The rapid growth of the Internet of Things (IoT) has revolutionized technology interaction, facilitating seamless connectivity among various smart devices and offering unprecedented opportunities for automation and data-driven decision-making [1]. However, this IoT surge brings about significant challenges, including spectrum scarcity, interference management, and power efficiency [2].

Wireless communication plays a pivotal role in enabling real-time data exchange and collaboration among IoT devices. As the number of IoT devices continues to grow, traditional communication methods encounter limitations related to spectrum availability and efficiency

[3]. This necessitates innovative solutions capable of intelligently adapting to dynamic wireless environments and optimizing the utilization of limited spectral resources.

In today's increasingly wireless-dependent world, cognitive radio offers numerous advantages. It optimizes the use of available radio frequencies, preventing congestion and interference. Moreover, it provides greater flexibility, enabling devices to adapt to changing conditions in their radio environment [6][7].

Cognitive radio finds applications in various fields, including wireless networks, mobile communications, sensor networks, and the Internet of Things (IoT) [8]. It promotes device interoperability and facilitates a more efficient use of limited radio resources [9].

A crucial component of cognitive communication is the antenna system, serving as the gateway for transmitting and receiving data [10][11]. The design of an antenna is influenced by its intended usage and the surrounding environment during system operation [12]. Developing antennas for IoT applications requires a thorough analysis of size, bandwidth, radiation pattern, and polarization. These factors significantly impact the antenna's performance and its adaptability to dynamic environments [13].

Miniaturized antennas are compact, physically small antenna structures designed to operate in confined spaces and low-profile devices while maintaining acceptable performance characteristics [14]. They find various applications in IoT, such as wearable devices, sensors, actuators, and industrial IoT applications like smart factories and smart buildings [15].

Several techniques are employed to miniaturize antennas. One common method involves geometric modifications to the antenna patch and ground plane, achieved by altering the antenna's shape or adding slots or

notches [16]. Alternatively, antennas can be miniaturized using metamaterial structures. Metamaterials are artificial materials with unique properties not found in natural materials, enabling the design of significantly smaller antennas compared to traditional ones [17]. However, introducing electrical shorts between the ground plane and the radiating element is another approach, but it may lead to undesirable effects such as increased insertion loss, reduced bandwidth, and signal quality deterioration [18].

In this article, we introduce a miniaturized antenna designed for cognitive radio applications in IoT. The antenna is designed and simulated using CST Studio Suite (Computer Simulation Technology) to operate at 5.8 GHz, the current standard frequency for IoT communications. Geometric modifications are applied to miniaturize the antenna. This technique was selected for its simplicity and versatility, as it can be implemented across a range of antenna shapes, making installation and integration into IoT devices more straightforward.

II. RECTANGULAR PATCH ANTENNA DESIGN

Our study consists primarily in proposing a rectangular patch antenna operating at the 5.8 GHz frequency, which is widely used in the IoT field. This antenna must satisfy the following specifications:

- Relative permittivity of the substrate $\epsilon_r=4.3$ Type FR4.
- Dielectric substrate height $h=1.6\text{mm}$.
- Input impedance $Z_c = 50\Omega$.

Our basic structure below (Figure 1) consists of a patch antenna printed on a dielectric substrate made of FR4 material, a ground plane and a feed with a microstrip line. To adjust the frequency and match the antenna to the desired frequency at 5.8 GHz, notches are inserted into the microstrip line.

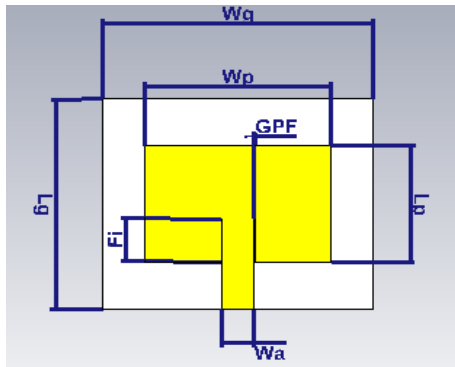


Fig.1 Geometry of antenna

The following table indicates the parameters of the initial patch antenna:

TABLE I. DIMENSIONS OF ANTENNA

Parameters	Dimensions (mm)
Patch width (W_p)	17.6
Patch length (L_p)	11.88
Patch height (t)	0.035
Substrate width (W_g)	27.48
Substrate length (L_g)	21.48
Substrate height (h)	1.6
Gap width (gpf)	0.05
Gap length (fi)	4.38
Microstrip feed length (L_a)	9.18
Microstrip feed width (W_a)	3.137

III. RESULTS AND DISCUSS

The designed antenna was simulated with three-dimensional simulator tool (CST). The result of the simulations is presented in this section.

The reflection coefficient (S_{11}) is generally used to characterize antenna performance. A low S_{11} value indicates a good impedance match between the antenna and the system. As shown in Figure 2, we have a multiband antenna with a good match of $S_{11} = -20.58 \text{ dB}$ at the desired frequency $f=5.8 \text{ GHz}$ and $S_{11} = -13 \text{ dB}$ at the frequency 7.9 GHz .

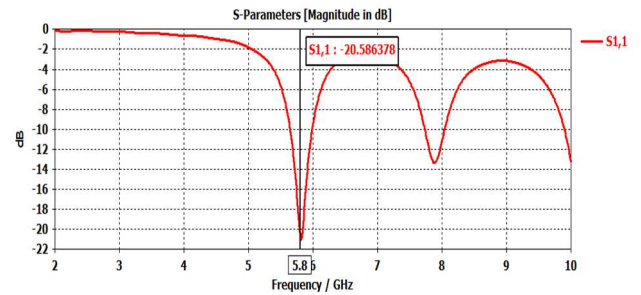


Fig.2 The reflection coefficient of the rectangular patch antenna

Figure 3 shows the voltage standing wave ratio (VSWR) with a value of 1.20, which indicates a good impedance match between the antenna and the feed line, as it is in the range of 1 to 2.

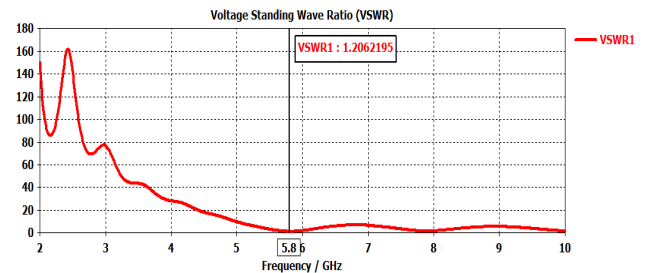


Fig.3 VSWR of the rectangular patch antenna

Figure 4 illustrates the gain variation as a function of frequency. At 5.8 GHz, we obtained a good gain of about 2.55 dBi. These results demonstrate the efficiency of this antenna.

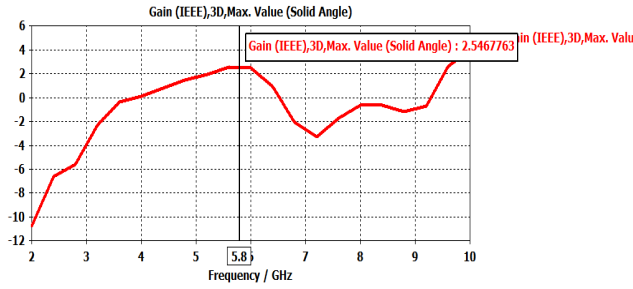


Fig.4 The gain of the rectangular patch antenna

The 2D and 3D radiation patterns of the antenna at 5.8 GHz are shown in Figures 5,6 and 7. The radiation pattern of the antenna is practically omnidirectional in both the H($\phi=90$) and E($\phi=0$) planes. It has a directivity of 5.94 dBi and a 3 dB opening angle of 94 degrees in the E($\phi=0$) plane and 98.6 degrees in the H($\phi=90$) plane.

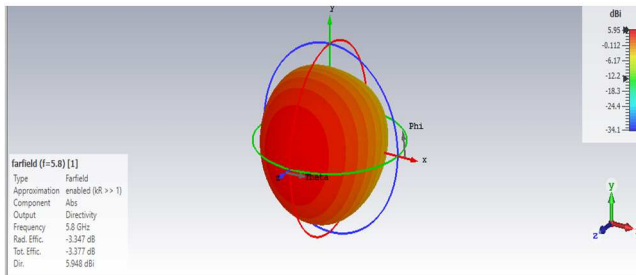


Fig.5 3D radiation diagram of the rectangular patch antenna

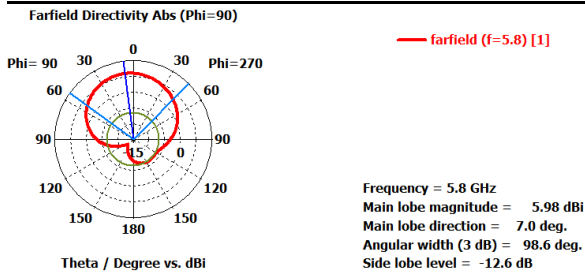


Fig.6 Polar radiation pattern of the rectangular patch antenna in H plane

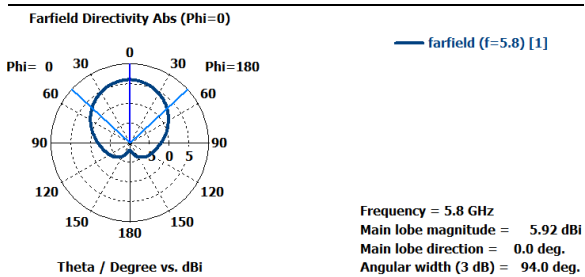


Fig.7 Polar radiation pattern of the rectangular patch antenna in E plane

IV. PATCH ANTENNA MINIATURIZATION

Nowadays, antennas are used more and more in a variety of fields that require scale models. These new constraints force us to miniaturize them for integration at different frequencies.

There are several miniaturization techniques, and in this section we'll choose to miniaturize our antenna by modifying the antenna geometry, as this technique is currently the most widely used. It consists of adding slots to the patch and ground.

A. On the patch

Figure 8 is the new antenna structure after the patch modifications. We've added circles at the ends with a radius of 3.25mm and rectangles in the middle with dimensions of 4mm x 10mm.

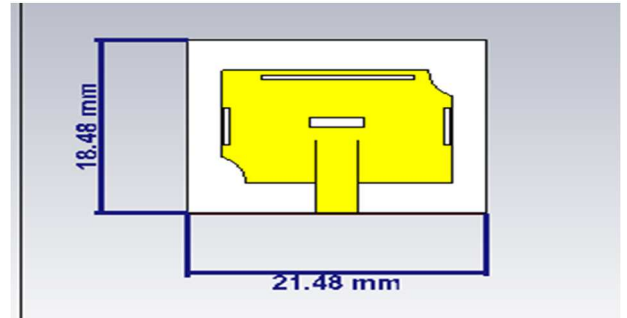


Fig.8 Miniaturized antenna on the patch

B. On the ground

Modifications can also be made to reduce the overall size of an antenna. By introducing slots in the ground plane, the effective electrical length of an antenna can be increased. This makes it possible to achieve the same performance as a larger antenna without occupying the same physical space.

In our case (Figure 9), we'll add two parallel rectangles to the right and left of 3mm with a width of 2.25mm and two other parallel rectangles of 6.5mm with a width of 2.5mm.

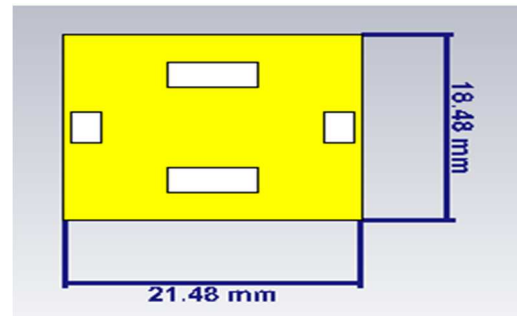


Fig.9 Miniaturized antenna on the ground

The new dimensions of the miniaturized patch antenna are shown in the following table:

TABLE II. THE NEW DIMENSIONS OF MINIATURIZED ANTENNA

Parameters	Dimensions (mm)
W_g	21.48
L_g	18.48
W_p	16.6
L_p	11.88
Miniaturization rate = 27%	

Figure 10 shows the reflection coefficient S_{11} with a multiband of -16.36dB at the resonant frequency of 5.8 GHz with a frequency bandwidth varying from 5.69 to 5.96 GHz and -13dB at the resonant frequency of 8.5 GHz. This is a satisfactory result for our study.

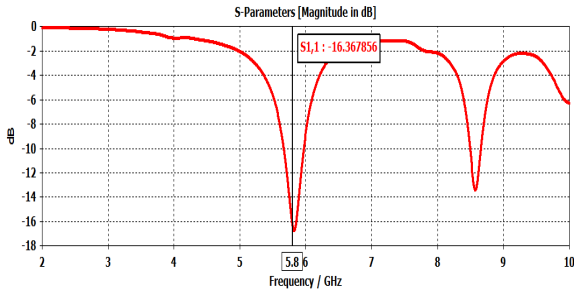


Fig.10 The reflection coefficient of miniaturized antenna

Figure 11 below illustrates the Standing Wave Ratio with a value of 1.358 less than 2, indicating good impedance matching between the antenna and the transmission line.

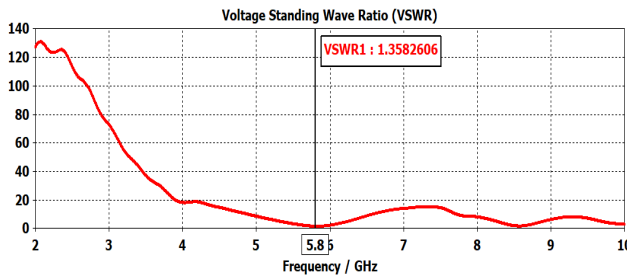


Fig.11 VSWR of miniaturized antenna

According to Figure 12 below, the gain is equal to 1.98dBi. This result corresponds to the expectations of our study.

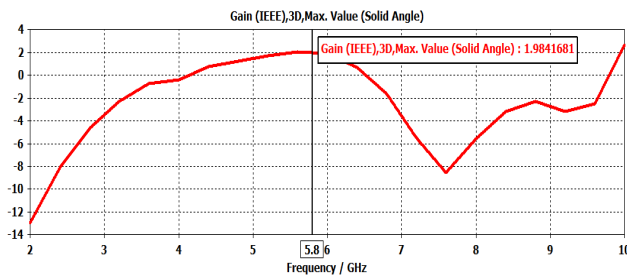


Fig.12 The gain of miniaturized antenna

The radiation pattern of the antenna in 2D and 3D is shown in Figures 13,14,15 below. We can see that the antenna has a quasi-omnidirectional radiation pattern in the E(phi=0) and H(phi=90) planes, with a directivity of 5.97dBi, a 3 dB aperture angle of 95.1 degrees for the E(phi=0) plane and 96.5 degrees for the H(phi=90) plane.

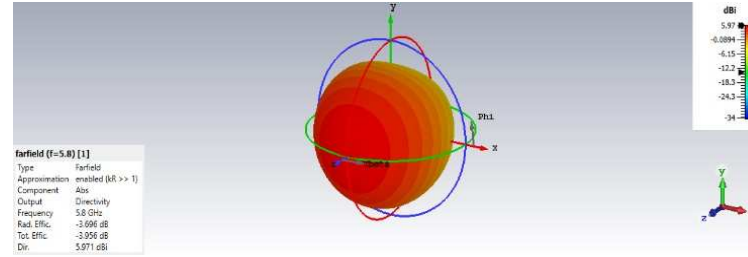


Fig.13 3D radiation diagram of miniaturized antenna

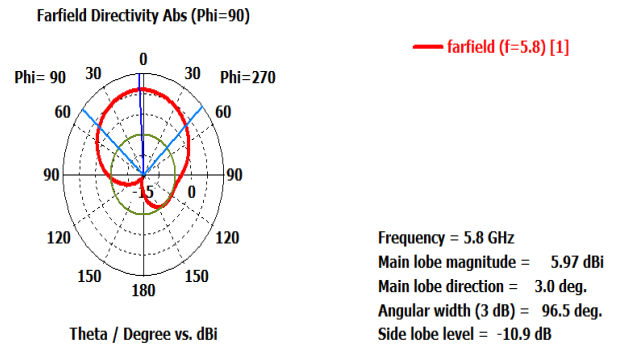


Fig.14 Polar radiation of miniaturized antenna in H plane

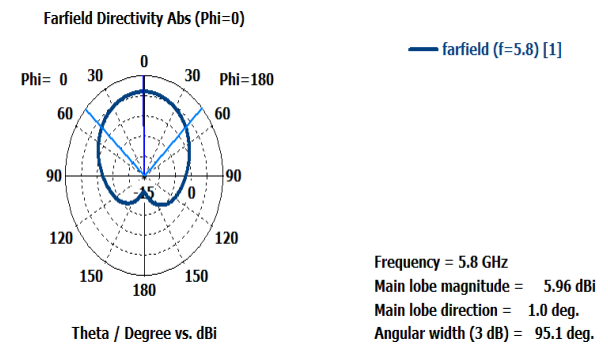


Fig.15 Polar radiation of miniaturized antenna in E plane

V. CONCLUSION

This paper presents a comprehensive study on a miniaturized antenna designed for cognitive communications in IoT applications. Using CST software, the antenna's performance was optimized at a 5.8 GHz center frequency. The iterative design approach refined parameters for optimal gain, directivity, and efficiency. The compact size with a 27% miniaturization rate makes it a promising choice for IoT devices with critical space and power constraints. The antenna's excellent performance indicates its potential to enhance cognitive communications and contribute to IoT technology advancement.

Future work may involve practical prototype development and field testing for real-world validation.

REFERENCES

- [1] Y. Khan, M. B. M. Su'ud, M. M. Alam, S. F. Ahmad, A. Y. A. B. Ahmad (Ayassrah), and N. Khan, "Application of Internet of Things (IoT) in Sustainable Supply Chain Management," *Sustainability*, vol. 15, no. 1, p. 694, Dec. 2022.
- [2] F. F. Qureshi, R. Iqbal, and M. N. Asghar, "Energy efficient wireless communication technique based on cognitive radio for internet of things," *Journal of Network and Computer Applications*, vol. 89, pp. 14–25, 2017.
- [3] V. Jahmunah et al., "Future IoT tools for COVID - 19 contact tracing and prediction: A review of the state - of - the - science," *International Journal of Imaging Systems and Technology*, vol. 31, no. 2, pp. 455–471, Feb. 2021.
- [4] M. Z. Baba-Ahmed, S. Tahraoui, A. Sedjelmaci, M. Bouregaa, and M. A. Rabah, "Self-management of autonomous agents dedicated to Cognitive Radio Networks," *Lecture Notes in Networks and Systems*, pp. 372–380, 2019.
- [5] G. Eappen, T. Shankar, and R. Nilavalan, "Advanced squirrel algorithm - trained neural network for efficient spectrum sensing in cognitive radio - based air traffic control application," vol. 15, no. 10, pp. 1326–1351, Feb. 2021.
- [6] M. Khasawneh, A. Azab, S. Alrabaei, H. Sakkal, and H. H. Bakhit, "Convergence of IOT and Cognitive Radio Networks: A survey of applications, techniques, and challenges," *IEEE Access*, vol. 11, pp. 71097–71112, 2023.
- [7] M. Z. Baba-Ahmed, B. Benmammar, and F.T. Bendimrad, "Spectrum Allocation for Autonomous Cognitive Radio Networks," *International Journal of Advancements in Computing Technology*, vol. 7, no 2, p. 48, 2015.
- [8] R. Prajapat, R. N. Yadav, and R. Misra, "Energy-Efficient k-Hop Clustering in Cognitive Radio Sensor Network for Internet of Things," *IEEE Internet of Things Journal*, vol. 8, no. 17, pp. 13593–13607, Sep. 2021.
- [9] A. J. Onumanyi, A. M. Abu-Mahfouz, and G. P. Hancke, "Low Power Wide Area Network, Cognitive Radio and the Internet of Things: Potentials for Integration," *Sensors*, vol. 20, no. 23, p. 6837, Nov. 2020.
- [10] F. Firouzi, B. Farahani, and A. Marinšek, "The convergence and interplay of edge, fog, and cloud in the AI-driven Internet of Things (IoT)," *Information Systems*, p. 101840, Jul. 2021.
- [11] R. D. Taleb, Mohammed Zakarya Baba-Ahmed, F. Bousalah, and M. Rabah, "Reconfigurable and Ecological Intelligent Antenna for Satellite Communication," In: Hatti, M. (eds) *Advanced Computational Techniques for Renewable Energy Systems. IC-AIRES 2022. Lecture Notes in Networks and Systems*, Springer, pp. 183–188, Jan. 2023.
- [12] M. Z. Baba-Ahmed, M.R.E.M. Fekar, N. Seladji-Hassaine and F.Z. Marouf, "Analysis of Patch and Printed Antennas integrated into UHF RFID Passive Tags," *International Journal Of Microwave and Optical Technology*, vol. 11, no. 6, pp. 460–468, Nov. 2016.
- [13] Md. A. Ullah, R. Keshavarz, M. Abolhasan, J. Lipman, K. P. Esselle, and N. Shariati, "A Review on Antenna Technologies for Ambient RF Energy Harvesting and Wireless Power Transfer: Designs, Challenges and Applications," *IEEE Access*, vol. 10, pp. 17231–17267, 2022.
- [14] S. Chourasia, S. K. Sharma, and D. P. Goswami, "Review on Miniaturization Techniques of Microstrip Patch Antenna," *SSRN Electronic Journal*, 2020.
- [15] A. A. Elijah and M. Mokayef, "Miniature microstrip antenna for IoT application," *Materials Today: Proceedings*, vol. 29, pp. 43–47, 2020.
- [16] N. Abbas, A. Basir, A. Iqbal, M. Yousaf, A. Akram, and H. Yoo, "Ultra-Miniaturized Antenna for Deeply Implanted Biomedical Devices," *IEEE Access*, vol. 10, pp. 54563–54571, 2022.
- [17] P. Kumar, T. Ali, and M. M. M. Pai, "Electromagnetic Metamaterials: A New Paradigm of Antenna Design," *IEEE Access*, vol. 9, pp. 18722–18751, 2021.
- [18] S. Joo, C. Lee, J. Kang, S. Seo, Y. Song, and J. Kim, "Intaglio Contact Printing of Versatile Carbon Nanotube Composites and Its Applications for Miniaturizing High - Performance Devices," *Small*, vol. 18, no. 3, p. 2106174, Dec. 2021.